

TELECOM CUSTOMER CHURN ANALYSIS DASHBOARD

1. Abstract

Customer churn is one of the most significant challenges faced by telecommunication companies. Churn occurs when customers discontinue using a company's services and switch to competitors. This project focuses on analyzing customer churn patterns using Power BI to identify key factors influencing customer retention and customer attrition.

The dashboard provides interactive visualizations and insights into customer demographics, contract types, internet services, payment methods, revenue at risk, and customer tenure. By leveraging business intelligence techniques, the dashboard enables stakeholders to make data-driven decisions that can improve customer satisfaction and reduce churn rates.

2. Introduction

In today's highly competitive telecommunications industry, retaining existing customers is more cost-effective than acquiring new ones. Understanding the behavior of customers who are likely to leave can help organizations implement targeted retention strategies.

This project utilizes customer data from a telecom company and transforms it into an interactive Power BI dashboard. The dashboard highlights trends, patterns, and relationships between customer characteristics and churn behavior.

3. Objectives

The primary objectives of this project are:

- To analyze customer churn behavior.
- To identify factors contributing to customer attrition.
- To calculate overall churn rate.
- To determine revenue at risk due to customer churn.
- To analyze churn across different contract types.
- To evaluate the impact of internet services on churn.
- To provide actionable business insights through visualization.

4. Tools and Technologies Used

Tool	Purpose
Power BI Desktop	Data Visualization and Dashboard Development
Power Query	Data Cleaning and Transformation
DAX	Calculated Measures and KPIs
Excel/CSV Dataset	Data Source

5. Dataset Description

The dataset contains customer information collected from a telecom service provider.

Key Attributes

- Customer ID
- Gender
- Senior Citizen Status
- Contract Type
- Internet Service Type
- Payment Method
- Monthly Charges
- Tenure
- Churn Status

The dataset consists of approximately 7,000 customer records.

6. Dashboard Overview

The dashboard consists of the following components:

KPI Cards

Total Customers

Displays the total number of customers currently available in the dataset.

Churn Rate

Shows the percentage of customers who have discontinued services.

Revenue at Risk

Represents potential revenue loss due to churned customers.

Average Tenure

Displays the average duration customers remain with the company.

Interactive Filters

The dashboard includes slicers for:

- Contract Type
- Internet Service
- Payment Method
- Gender
- Senior Citizen
- Charge Group

These filters allow dynamic exploration of customer behavior.

7. Visual Analysis

Customer Churn Distribution

The donut chart presents the proportion of churned and retained customers.

Observation:

- Approximately 26.54% of customers have churned.
- Around 73.46% remain active.

Contract Type Analysis

The stacked bar chart compares churn across contract categories.

Observation:

- Month-to-month contracts exhibit the highest churn.
- One-year and two-year contracts demonstrate better customer retention.

Business Insight: Long-term contracts significantly reduce churn probability.

Internet Service Analysis

The clustered column chart evaluates churn by internet service type.

Observation:

- Fiber optic customers show the highest churn rate.
- DSL customers exhibit moderate churn.

- Customers without internet services have the lowest churn.

Business Insight: Service quality improvements for fiber optic users may reduce churn.

Charge Group Analysis

The line chart illustrates churn across different charge groups.

Observation:

- Customers with higher monthly charges show increased churn tendencies.
- Lower charge groups exhibit relatively lower churn rates.

Business Insight: Pricing strategies should be reviewed for customers paying premium charges.

8. Key Findings

- Overall churn rate is 26.54%.
- Month-to-month contracts are the most vulnerable segment.
- Fiber optic service users have higher churn tendencies.
- High-paying customers contribute significantly to revenue at risk.
- Customers with longer tenure demonstrate greater loyalty.
- Contract duration is a strong predictor of customer retention.

9. Recommendations

Based on the analysis, the following recommendations are proposed:

1. Encourage customers to migrate to long-term contracts.
2. Introduce loyalty and retention programs for high-risk customers.
3. Improve customer support for fiber optic users.
4. Offer personalized discounts to customers with high monthly charges.
5. Monitor newly acquired customers closely during early tenure periods.
6. Implement predictive churn models to proactively identify at-risk customers.

10. Conclusion

The Telecom Customer Churn Analysis Dashboard successfully provides meaningful insights into customer retention and churn behavior. Through interactive visualizations and KPI monitoring, decision-makers can identify high-risk customer segments and implement effective retention strategies.

The dashboard serves as a valuable business intelligence solution that supports data-driven decision-making, improves customer satisfaction, and helps reduce revenue loss caused by customer churn.

